PHYSICS FORMULA & METHODS HANDBOOK

CHAPTER 5: WORK, ENERGY AND POWER | CBSE / JEE / NEET EXAM-ORIENTED NOTES

1. Scalar Product of Vectors

The scalar (dot) product of two vectors $\vec{A}$ and $\vec{B}$ yields a scalar quantity[cite: 52].

TOPIC: SCALAR PRODUCT DEFINITION

$$\vec{A} \cdot \vec{B} = AB\cos\theta$$

Symbols: θ = angle between vectors, A, B = magnitudes.
Component Form: $\vec{A} \cdot \vec{B} = A_xB_x + A_yB_y + A_zB_z$[cite: 52]
Condition for Perpendicularity: $\vec{A} \cdot \vec{B} = 0$ (since $\cos 90^\circ = 0$)[cite: 52]

MUST KNOW

Unit vector dot products: $\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1$ and $\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$[cite: 52].

2. Work & The Work-Energy Theorem

TOPIC: WORK DONE BY CONSTANT FORCE

$$W = (F\cos\theta)d = \vec{F} \cdot \vec{d}$$

Symbols: F = force magnitude, d = displacement, θ = angle between force and displacement.
SI Unit: Joule (J) or $\mathrm{kg\ m^2 s^{-2}}$. Dimensions: $[ML^2T^{-2}]$[cite: 54].

TOPIC: WORK DONE BY VARIABLE FORCE (1D)

$$W = \int_{x_i}^{x_f} F(x)dx = \text{Area under } F\text{-}x \text{ curve}$$

TOPIC: WORK-ENERGY THEOREM

$$W_{\text{net}} = K_f - K_i = \Delta K$$

Statement: The work done by the net force on an object equals the change in its kinetic energy[cite: 53].
Kinetic Energy: $K = \frac{1}{2}mv^2$[cite: 54]

CAUTION

Work done by friction or viscous drag is generally negative ($\theta = 180^\circ$), reducing the kinetic energy of the system[cite: 54].

3. Potential Energy & Conservative Forces

TOPIC: CONSERVATIVE FORCE & POTENTIAL ENERGY RELATION

$$F(x) = -\frac{dV(x)}{dx} \implies \Delta V = -W_c = -\int_{x_i}^{x_f} F(x)dx$$

TOPIC: GRAVITATIONAL & SPRING POTENTIAL ENERGY

$$V_{\text{grav}} = mgh \quad \text{and} \quad V_{\text{spring}} = \frac{1}{2}kx^2$$

Hooke's Law (Spring Force): $F_s = -kx$ (where k is spring constant in N m^{-1}).

4. Conservation of Mechanical Energy

TOPIC: TOTAL MECHANICAL ENERGY (E)

$$E = K + V = \text{constant} \quad (\text{when only conservative forces act})$$

Non-Conservative Forces Present: $E_f - E_i = W_{\text{nc}}$ (where W_{nc} is work done by friction/external non-conservative forces).

5. Power

TOPIC: INSTANTANEOUS & AVERAGE POWER

$$P = \frac{dW}{dt} = \vec{F} \cdot \vec{v} \quad \text{and} \quad P_{\text{av}} = \frac{W}{t}$$

SI Unit: Watt (W) or J s^{-1}. Dimensions: $[ML^2T^{-3}]$.
Conversion: $1\ \text{hp} = 746\ \text{W}$.

6. Collisions

TOPIC: ONE-DIMENSIONAL ELASTIC COLLISION (HEAD-ON)

$$v_{1f} = \frac{m_1 - m_2}{m_1 + m_2} v_{1i} \quad \text{and} \quad v_{2f} = \frac{2m_1 v_{1i}}{m_1 + m_2}$$

Key Principle: Total linear momentum is conserved in all collisions. Kinetic energy is conserved only in elastic collisions.
Equal Masses ($m_1 = m_2$): $v_{1f} = 0,\ v_{2f} = v_{1i}$ (Velocities are exchanged).

EXAM FAVOURITE

For a 2D elastic collision between two equal masses ($m_1 = m_2$) with one stationary target, the two bodies move at **right angles** to each other after collision ($\theta_1 + \theta_2 = 90^\circ$).

7. Energy Units & Conversions

Unit Name	Symbol / Value in Joules
Erg	$1\ \text{erg} = 10^{-7}\ \text{J}$
Electron Volt	$1\ \text{eV} = 1.6 \times 10^{-19}\ \text{J}$
Calorie	$1\ \text{cal} = 4.186\ \text{J}$
Kilowatt-hour	$1\ \text{kWh} = 3.6 \times 10^{6}\ \text{J}$